

CSCI 5408

Data Management, Warehousing Analytics

Assignment Report

Class Assignment -2

Banner ID: B00948964

Gitlab Repository Link:

https://git.cs.dal.ca/pgajera/csci5408_f23_b00948964_parth_gajera

Problem 1A: Reuter News Data Reading & Transformation and storing in MongoDB.

Here in this task we have to do the following two operation, first is the data reading and second is transforming data and then at last we have to store the data in the MongoDB in the form of documents.

Here I have created the MongoDB instance in the MongoDB compass app. And then connected to my instance in the compass app.

Algorithm:

Main Method Execution:

- 1.1. Define the MongoDB connection string.
- 1.2. Create a MongoClientURI object using the connection string.
- 1.3. Create a MongoClient using the MongoClientURI.
- 1.4. Access the MongoDB database.
- 1.5. Get the MongoDB collection named "news".
- 1.6. Specify an array of Reuters news files.
- 1.7. Iterate through each news file and process it using the processFile method.
- 1.8. Close the MongoClient when done.

Process File Method:

- 2.1. Open a BufferedReader for reading the Reuters news file.
- 2.2. Initialize a StringBuilder to store a block of Reuters content.
- 2.3. Iterate through each line of the file.
- 2.4. If a line contains the start of a Reuters block (<REUTERS), clear the StringBuilder.
- 2.5. Append each line to the StringBuilder.
- 2.6. If a line contains the end of a Reuters block (</REUTERS>):
 - Extract the values of <TITLE> and <BODY> using the extractValue method.
 - Check if both title and body are not null before inserting into MongoDB.
 - Insert a new document with title and body into the MongoDB collection.
 - Optionally, log a warning if the body is null but the title is not.
- 2.7. Close the BufferedReader when done.

Extract Value Method:

3.1. Given a block of text and a tag name, find the start and end indices of the tag.

3.2. If the tag is found and indices are within bounds:

- Extract and return the value between the start and end indices.

3.3. If the tag is not found or indices are out of bounds:

- Return null.

Flowchart:

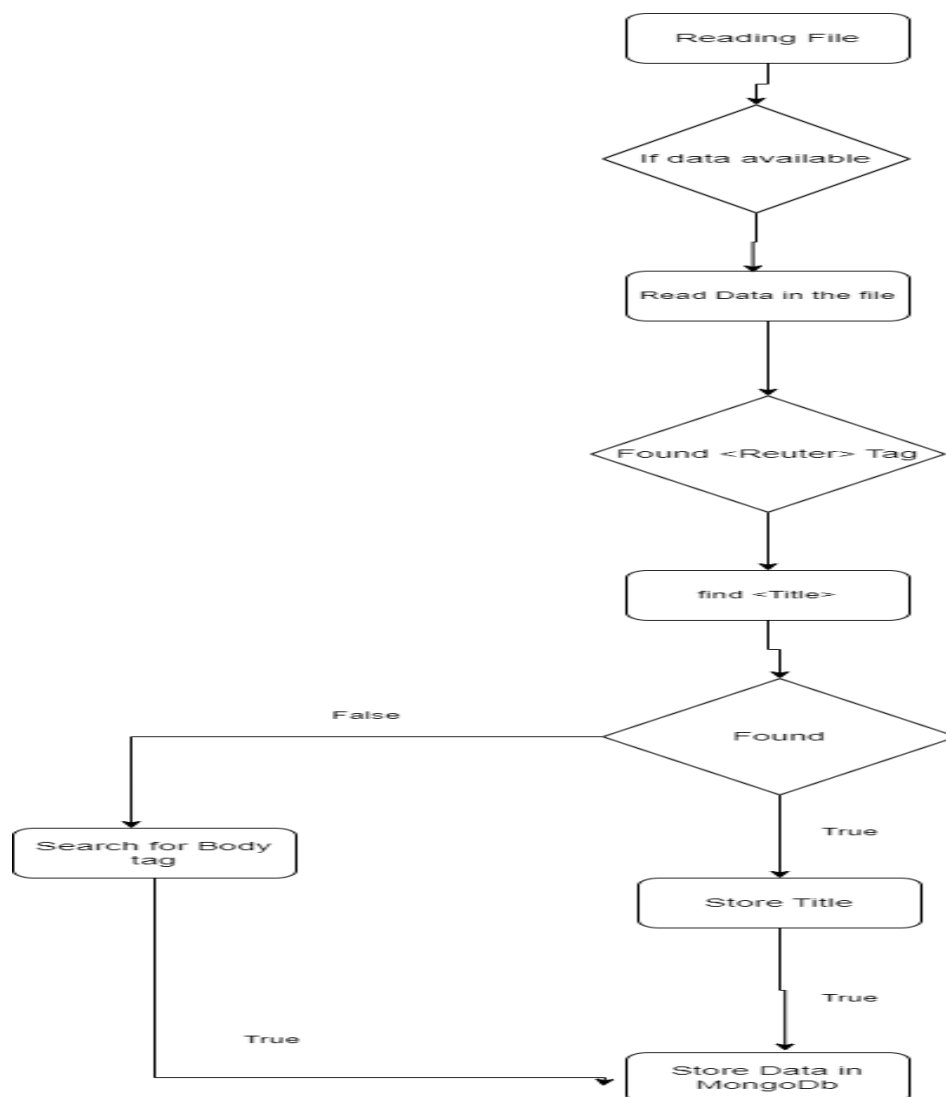
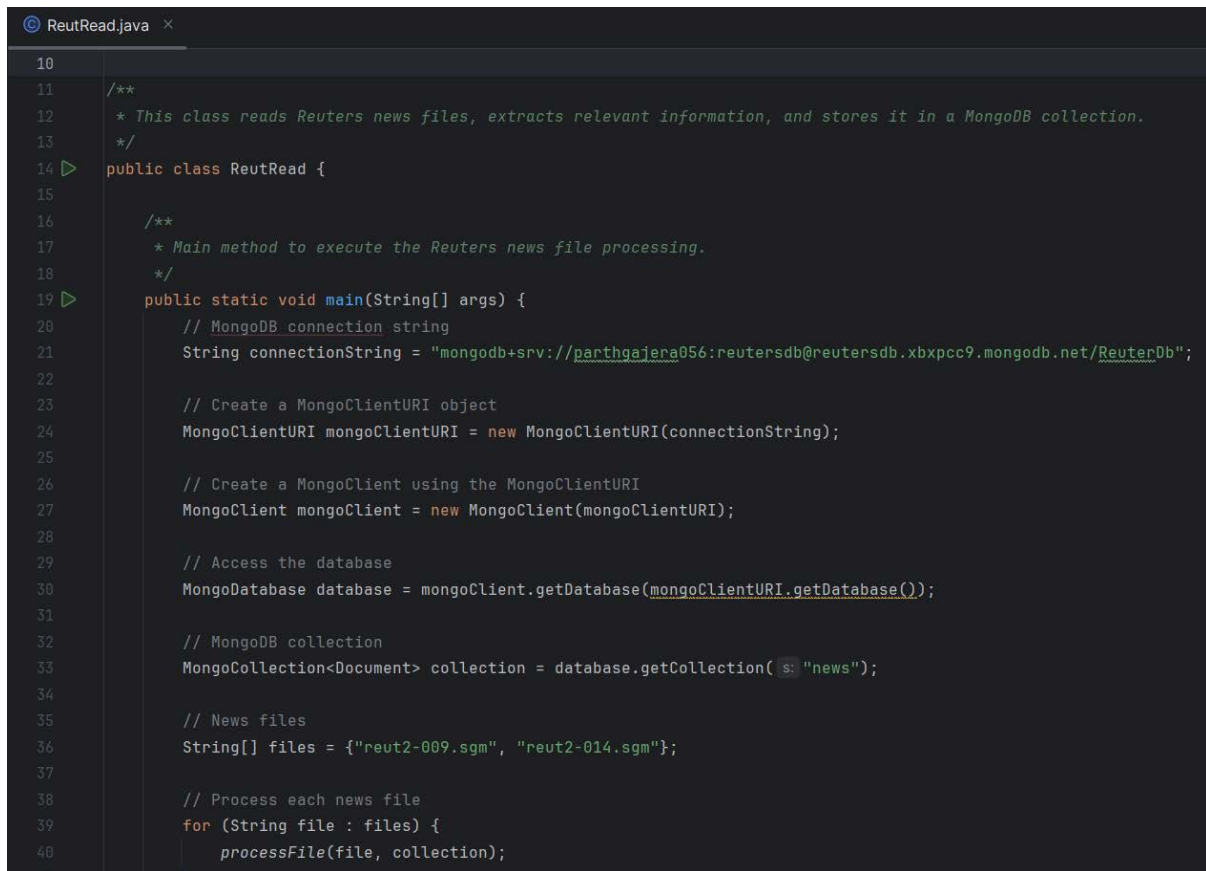


Figure 1: Flowchart

Below is the snapshot of the code for the data reading and transformation and then storing in the MongoDB:



```
10
11 /**
12  * This class reads Reuters news files, extracts relevant information, and stores it in a MongoDB collection.
13  */
14 public class ReutRead {
15
16     /**
17      * Main method to execute the Reuters news file processing.
18      */
19     public static void main(String[] args) {
20         // MongoDB connection string
21         String connectionString = "mongodb+srv://parthgajera056:reutersdb@reutersdb.xbxpc9.mongodb.net/ReuterDb";
22
23         // Create a MongoClientURI object
24         MongoClientURI mongoClientURI = new MongoClientURI(connectionString);
25
26         // Create a MongoClient using the MongoClientURI
27         MongoClient mongoClient = new MongoClient(mongoClientURI);
28
29         // Access the database
30         MongoDB database = mongoClient.getDatabase(mongoClientURI.getDatabase());
31
32         // MongoDB collection
33         MongoCollection<Document> collection = database.getCollection("news");
34
35         // News files
36         String[] files = {"reut2-009.sgm", "reut2-014.sgm"};
37
38         // Process each news file
39         for (String file : files) {
40             processFile(file, collection);
41         }
42     }
43 }
```

Figure 2: ReutRead.java

Now I will show the output of the above code, the data stored in the mongodb as a document. Here the title and the body tag are saved in the form of document in the database which is shown in the picture:

ReuterDb.news

[Documents](#) [Aggregations](#) [Schema](#) [Indexes](#) [Validation](#)

Filter 

Type a query: { field: 'value' } or [Generate query](#) 

[Explain](#) [Reset](#)

[+ ADD DATA](#) [EXPORT DATA](#)

1 - 20 of 1694 

```
_id: ObjectId('656a5931a09a41684d4b00e3')
title: "ADVANCED MAGNETICS &lt;ADMG> IN AGREEMENT"
body: "Advanced Magnetis Inc said it
      reached a four mln dlrs research and de..."
```

```
_id: ObjectId('656a5933a09a41684d4b00e4')
title: "HEALTH RESEARCH FILES FOR BANKRUPTCY"
body: "&lt;Health Research and
      Management Group> said it has filed for protec..."
```

```
_id: ObjectId('656a5933a09a41684d4b00e5')
title: "NUMEREX CORP &lt;NMRX> 2ND QTR JAN 31 LOSS"
body: "Shr loss seven cts vs profit five cts
      Net loss 149,421 vs profit 1..."
```

```
_id: ObjectId('656a5934a09a41684d4b00e6')
title: "U.S. SELLING 12.8 BILLION DLRS OF 3 AND 6-MO BILLS MARCH 30 TO PAY DOW..."
body: null
```

```
_id: ObjectId('656a5934a09a41684d4b00e7')
```

Figure 3: Document Geneated in MongoDB

Here as seen in the picture the total document generated is 1697 and inserted in the document.

Problem 1B: Reuter News Data Reading & Transformation and storing in MongoDB.

First, I have made an instance on GCP by creating Cloud Dataproc cluster on the GCP.

The below is the snapshot for the spark code written in java for the word analyzer:

```
ReutersWordAnalyzer.java x
10 import java.nio.file.Files;
11 import java.util.Arrays;
12 import java.util.HashSet;
13 import java.util.List;
14 import java.util.Set;
15
16 /**
17  * Spark application for analyzing word frequencies and identifying unique words in Reuters news articles.
18  */
19 public class ReutersWordAnalyzer {
20
21     // Set of common stop words to filter out during word count
22     private static final Set<String> STOP_WORDS = new HashSet<>(Arrays.asList(
23         "a", "an", "and", "are", "as", "at", "be", "but", "by", "for", "if", "in",
24         "into", "is", "it", "no", "not", "of", "on", "or", "such", "that", "the",
25         "their", "then", "there", "these", "they", "this", "to", "was", "will", "with"
26     ));
27
28     /**
29      * Main method to execute the ReutersWordAnalyzer.
30      */
31     public static void main(String[] args) {
32         // Path to the Reuters news file
33         String reutersFile = "reut2-009.sgm";
34
35         try {
36             // Set up Spark
37             SparkSession spark = SparkSession.builder().appName("ReutersWordAnalyzer").getOrCreate();
38             JavaSparkContext sc = new JavaSparkContext(spark.sparkContext());
39
40             // Read and preprocess the Reuters news file
```

Figure 4: ReuterWordAnalyzer.java

Problem-2: Sentiment Analysis using BOW model on title of Reuters News Articles.

Here in this task, we have to perform sentiment analysis using BOW (Bag of Word) model.

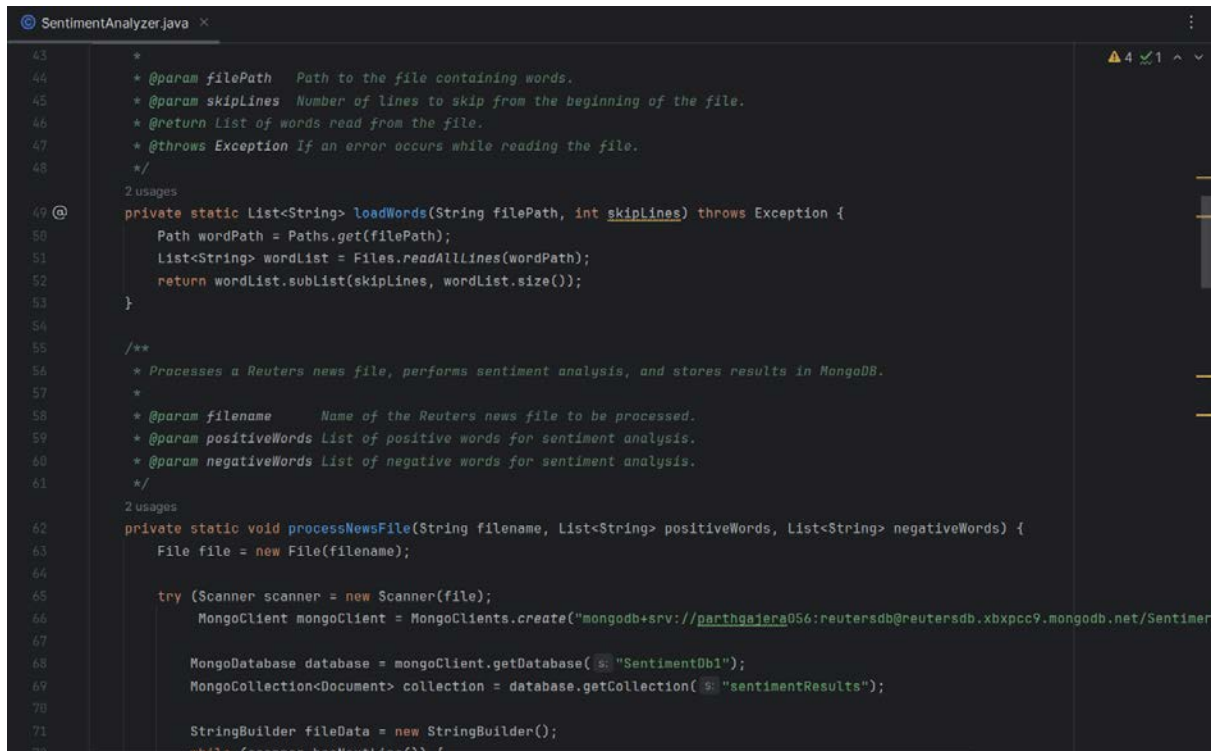
Here I write a java code which will iterate over the collection and will generate the bag of words and I have used MongoDB database to store it and then I exported it into a CSV file.

Here I Fill positive and negative bags based on positive and negative word dataset.

Calculate score based on positive and negative numbers. +1 is given when positive word is found and -1 is given for negative word.

Determine the polarity of news title based on the score.

The code snippet is shown below:



```
43  *
44  * @param filePath Path to the file containing words.
45  * @param skipLines Number of lines to skip from the beginning of the file.
46  * @return List of words read from the file.
47  * @throws Exception If an error occurs while reading the file.
48  */
49  2 usages
50  @ private static List<String> loadWords(String filePath, int skipLines) throws Exception {
51      Path wordPath = Paths.get(filePath);
52      List<String> wordList = Files.readAllLines(wordPath);
53      return wordList.subList(skipLines, wordList.size());
54  }
55  /**
56   * Processes a Reuters news file, performs sentiment analysis, and stores results in MongoDB.
57   *
58   * @param filename Name of the Reuters news file to be processed.
59   * @param positiveWords List of positive words for sentiment analysis.
60   * @param negativeWords List of negative words for sentiment analysis.
61   */
62  2 usages
63  private static void processNewsFile(String filename, List<String> positiveWords, List<String> negativeWords) {
64      File file = new File(filename);
65
66      try {Scanner scanner = new Scanner(file);
67          MongoClient mongoClient = MongoClient.create("mongodb+srv://parthgajera056:reutersdb@reutersdb.xbxpc9.mongodb.net/SentimentAnalyzer?authSource=reutersdb");
68          MongoDBDatabase database = mongoClient.getDatabase("SentimentDb1");
69          MongoCollection<Document> collection = database.getCollection("sentimentResults");
70
71          StringBuilder fileData = new StringBuilder();
72          while (scanner.hasNextLine()) {
```

Figure 5: SentimentAnalyzer.java

I have used MongoDB to store the result in the collection name “sentimentResults” and the snapshot for that is shown below:

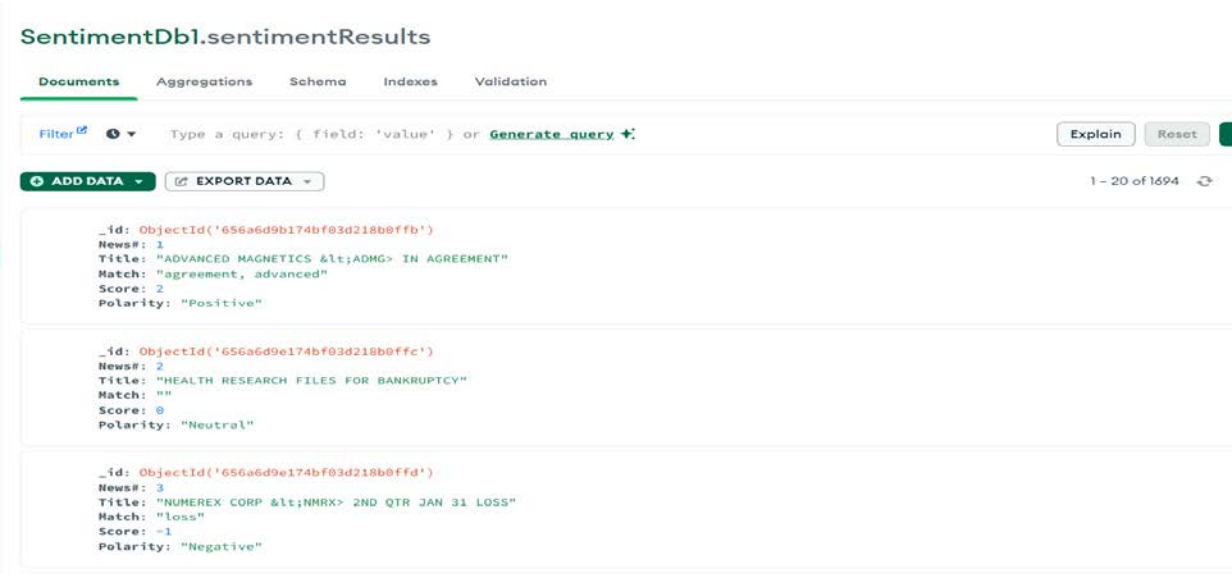


Figure 6: sentimentResults in MongoDB

Here is the snapshot of the CSV file which got generated after exporting from the MongoDB database:

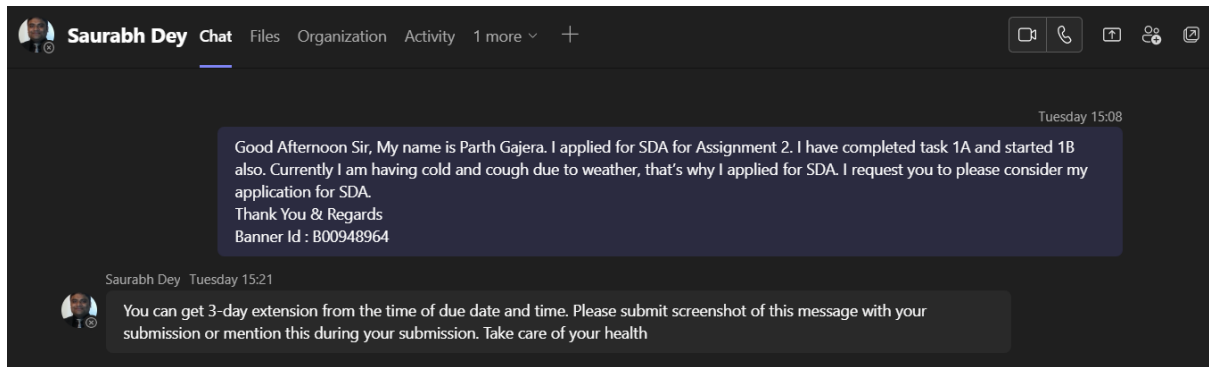
News#	Title	Match	Score	Polarity
1	ADVANCED MAGNETICS <ADMG> IN AGREEMENT	agreement, advanced	2	Positive
2	HEALTH RESEARCH FILES FOR BANKRUPTCY		0	Neutral
3	NUMEREX CORP <NMRX> 2ND QTR JAN 31 LOSS	loss	-1	Negative
4	U.S. SELLING 12.8 BILLION DLRS OF 3 AND 6-MO BILLS MARCH 30 TO PAY DOWN 1.2 BILLION DLRS		0	Neutral
5	U.S. 2-YEAR NOTE AVERAGE YIELD 6.43 PCT, STOP 6.44 PCT, AWARDED AT HIGH YIELD 85 PCT	awarded	1	Positive
6	COMMODORE <CBU>, ATARI IN SETTLEMENT		0	Neutral
7	BALDRIGE SUPPORTS NIC TALKS ON CURRENCIES	supports	1	Positive
8	TRIANGLE <TRI> BEGINS EXCHANGE OFFER		0	Neutral
9	SOUTHMARK <SM> UNIT IN PUBLIC OFFERING OF STOCK		0	Neutral
10	EASTMAN KODAK CO TO SELL HOLDINGS IN ICN PHARMACEUTICALS AND VIRATEK INC		0	Neutral
11	FEUD PERSISTS AT U.S. HOUSE BUDGET COMMITTEE		0	Neutral
12	TREASURY BALANCES AT FED ROSE ON MARCH 23		0	Neutral
13	FARM CREDIT SYSTEM SEEN NEEDING 800 MLN DLRS AID		0	Neutral
14	USX <X> USS UNIT RAISES PRICES		0	Neutral
15	UNIONIST URGES RETALIATION AGAINST JAPAN		0	Neutral

Figure 7: Sentiment Analysis Result.csv

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Assignment-2
CSCI 5408 Data Mgmt

Saurabh Sir's Confirmation for SDA:



References:

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- [2] “MongoDB: The Developer Data Platform | MongoDB”, MongoDB [Online]. Available: <https://www.mongodb.com/>. Accessed [Nov. 28, 2023].
- [3] “Cloud Computing Services | Google Cloud”, Google Cloud [Online]. Available: <https://cloud.google.com/?hl=en>. Accessed [Nov. 30, 2023].
- [4] “positive-words.txt”, gist.github.com [Online]. Available: <https://gist.github.com/mkulakowski/4289437>. Accessed [Nov. 28, 2023].
- [5] “negative-words.txt”, gist.github.com [Online]. Available: <https://gist.github.com/mkulakowski/4289441>. Accessed [Nov. 29, 2023].